

Table 1: Three intervention surfaces commonly called a “skill.”

Surface	What changes	Matched comparison
Persistent text	instruction in context	targeted vs. generic text
Supplied output	result placed in context	target vs. matched helper output
Executable state	hidden tool implementation	correct vs. matched procedure

IDENTIFYING CAUSAL SKILL EFFECTS: MATCHED EXECUTABLE INTERVENTIONS IN TIME-SERIES AGENTS

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ABSTRACT

Self-evolving agents often store reusable “skills,” yet evaluations can conflate instruction text, supplied procedure output and executable procedure state. We separate these intervention surfaces while keeping the downstream benchmark request fixed. Persistent text shows little matched separation, and prospective procedure-output injection does not improve downstream findings. Executable state produces larger matched effects. On seven first-time STL-decomposition endpoints, the correct hidden implementation succeeds on 5/7 tasks while an identical-interface helper succeeds on 0/7, giving +71.4 percentage points with exact one-sided $p = 0.03125$ under the frozen deterministic verifier. A deduplicated first-observation completion of the same pre-outcome candidate population gives 10/20 versus 2/20 ($p = 0.00390625$) as robustness evidence without new closure authority. Measurement robustness remains open: the only valid arm-blinded semantic reviewer judges 6/7 targeted answers and 2/7 generic answers fully correct, giving $p = 0.109375$; the second predeclared reviewer is nonvoting after a parse failure and is not replaced. We also test substantive task-local off-target procedures on previously unused L2 tasks spanning five non-STL families. Qwen2.5-32B gives 5/8 targeted successes versus 1/8 off-target (+50.0 points, $p = 0.0625$), while DeepSeek-v4-Pro gives 5/9 versus 2/9 (+33.3 points, $p = 0.125$). The non-STL directions are consistent across model families while both matched gates remain open. These results support matched executable interventions as a causal evaluation tool, while broader procedure-family generality, semantic measurement robustness and release-to-release persistent transfer remain open empirical questions.

1 INTRODUCTION

Suppose an agent analyzes a monthly energy series. In March it learns that a publication date must be aligned with the period represented by the report. The same operation is needed again in April after new evidence arrives. Calling the March experience a *skill* leaves an important ambiguity. The agent may store a sentence that reminds it what to do. It may store the March computation result. It may instead store an executable procedure that runs on the April evidence.

These objects can all change behavior, yet they support different causal claims. A reminder changes the language seen by the model. A supplied result changes the information placed in context. Executable state changes which computation becomes available after the model decides to call a tool. Aggregate before–after accuracy cannot identify which object carried the gain.

TimeSage-EV makes this distinction concrete. It evaluates 60 institutional scenarios over 1,485 scenario-period questions under changing evidence cutoffs (Yao et al., 2026). Its TimeSage-1.0 harness stores procedures that can become callable in later periods. The source paper reports higher

aggregate performance with this skill library and a token-cost ratio of $0.82 \times$ the reference condition. This establishes that persistent assistance can matter. It does not identify the causal object inside the library.

TimeSage-MT provides a complementary intervention substrate (Kong et al., 2026). Its task records contain visible data, benchmark procedures and pre-existing downstream verifiers. We use those records to change one skill surface at a time while keeping the benchmark request fixed.

Throughout the paper, an *endpoint* means one frozen benchmark turn together with its pre-existing verifier. A *matched control* receives the same visible tool interface as the targeted arm. Only the hidden implementation changes. The *primary gate* requires a paired risk-difference of at least +20 percentage points and one-sided exact $p \leq 0.05$. A support failure means that an experiment could not be executed; it carries no negative scientific authority.

The evidence ladder first tests weaker objects. Persistent targeted text gives 11/13 successes, compared with 10/13 for matched generic text. Prospective procedure-output injection gives 4/20 targeted successes and 5/20 matched-control successes. These results show that prompt persistence and answer-bearing context should not be counted as evidence for executable skill repair.

We next hide the intervention behind one native tool surface. The model sees the same function name, description and argument schema in both executable arms. The targeted implementation executes the benchmark-recorded procedure on hash-verified data, while the matched implementation changes only the hidden computation. On the initial Qwen executable set, targeted succeeds on 6/19 endpoints while a content-light matched helper succeeds on 2/19. A later first-time STL-decomposition tranche gives 5/7 targeted successes and 0/7 matched-helper successes, passing the frozen deterministic matched gate with exact $p = 0.03125$. A subsequent arm-blinded semantic measurement audit does not reproduce that inferential gate with its only valid reviewer: semantic correctness is 6/7 targeted versus 2/7 generic, with $p = 0.109375$. The second predeclared reviewer is nonvoting after a parse failure and is not replaced after labels are observed. We therefore preserve the frozen deterministic result while treating semantic measurement robustness as open.

The STL result raises a sharper question: does executable implementation value survive beyond one procedure family and a content-light control? We therefore construct a new population from TimeSage-MT L2 tasks that never entered any earlier candidate pool. The control now executes a different substantive benchmark procedure from the same task. Target and off-target outputs are word-count matched before model evaluation. Nine endpoints pass execution qualification and span five non-STL target procedure families. On Qwen2.5-32B, 8 endpoints are valid after one frozen protocol failure; targeted succeeds on 5/8 and off-target on 1/8, giving +50.0 points with $p = 0.0625$. The exact same nine-endpoint contract on DeepSeek-v4-Pro gives 5/9 versus 2/9, or +33.3 points with $p = 0.125$. The primary matched direction is positive in both model families, while neither broad non-STL gate closes.

Our contribution is therefore a matched-intervention evaluation standard for reusable skills together with a concrete empirical boundary. Executable implementation value has a frozen deterministic gate pass within STL decomposition, but semantic robustness of that pass remains unresolved. Stronger controls across five additional procedure families give positive directions on two model families without closing their matched gates. The remaining TimeSage-EV experiment adds chronology by writing a procedure before a later release and keeping that state fixed when future endpoints decide whether to call it.

2 RECURRING FAILURE AS A MISSING PROCEDURE

Let a scenario generate a sequence of release periods $t_1, \dots, t_K$. At each period, the agent receives a cutoff-valid evidence state E_t and a question x_t . The agent also carries a reusable skill library L_t . Future behavior follows

$$Y_t = H(x_t, E_t, L_t, S_t), \quad (1)$$

where S_t contains the remaining agent state.

A recurring failure mode appears when the same procedural error repeats across periods while the concrete evidence changes. This pattern suggests that the agent lacks a reusable operation. We represent the missing operation as a skill g_m associated with mechanism m .

Targeted temporal skills isolate a mechanism behind recurring grounding failures

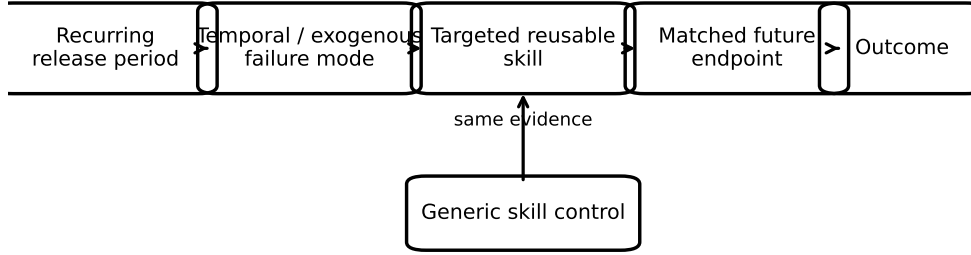


Figure 1: Targeted skill intervention. The same future endpoint is evaluated with a mechanism-specific skill and with a matched generic skill. This isolates the contribution of the procedure encoded by the targeted skill.

The causal estimand compares outcomes on the same future endpoint under two skill libraries:

$$\Delta_m = \mathbb{E}[Y_t \mid do(L_t = L \cup \{g_m\})] - \mathbb{E}[Y_t \mid do(L_t = L \cup \{g_c\})], \quad (2)$$

where g_c is a generic control skill with matched implementation complexity. A positive Δ_m supports the claim that the target procedure repairs the matched failure beyond generic skill assistance.

2.1 TEMPORAL CUTOFF SKILL

The cutoff skill receives a query date and a set of candidate evidence timestamps. It returns the admissible evidence subset under the benchmark cutoff. Its purpose is to prevent later releases from leaking into earlier-period analysis. The skill can also expose the cutoff date explicitly to subsequent reasoning.

2.2 RELEASE ALIGNMENT SKILL

The alignment skill maps source-release dates to the benchmark period represented by the evidence. It is useful when the source publication date differs from the underlying reporting period. The skill returns a normalized period identifier and verifies that the selected time-series row matches that period.

2.3 EXOGENOUS GROUNDING SKILL

The grounding skill links a numerical change to documentary evidence available before the cutoff. It searches prior-period reports for candidate events and returns evidence spans tied to the target period. This procedure supports attribution questions that require context outside the numerical series.

Each skill encodes a reusable operation that excludes scenario-specific answers. The same function can apply across future periods. This property matches the source harness contract for skills saved through `induct_skill`.

3 SOURCE EVIDENCE FOR THE SKILL BOTTLENECK

TimeSage-EV evaluates recurring analysis over institutional releases from February 2023 through May 2026 (Yao et al., 2026). The benchmark reports persistent gaps in temporal validity. It also reports recurring difficulty with exogenous context. These failures remain visible across model tiers, which makes them suitable targets for reusable procedure induction.

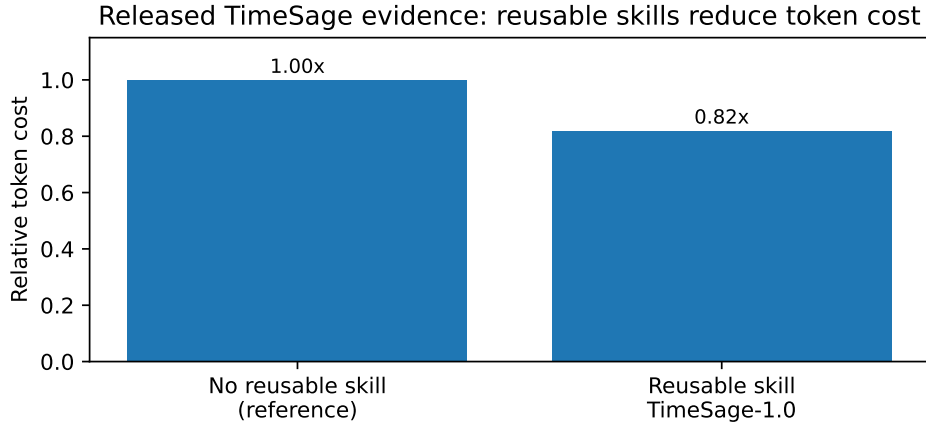


Figure 2: Released TimeSage evidence reports $0.82\times$ relative token cost when reusable skills are enabled. The source also reports higher benchmark performance under the skill-library condition.

The source paper introduces TimeSage-1.0 as a diagnostic self-evolving harness. The base interface remains aligned with the baseline. The skill-library manager is the additional mechanism. Skills are induced from previous reasoning traces and become available to later periods. This gives the benchmark an explicit state-update boundary for testing reusable procedures.

The released aggregate results support the value of this state. Reusable skills improve overall performance, with the clearest gains on medium and hard scenarios. Skill invocation is most frequent on hard scenarios. The source also reports that the skill-enabled condition reduces token cost to $0.82\times$ the reference condition.

These results establish three ingredients for the causal hypothesis. Reusable procedures can enter persistent state. They can influence later periods. They improve aggregate performance while reducing repeated reasoning cost. The remaining question concerns mechanism specificity: which recurring failure families are repaired by which skill?

The benchmark Appendix provides the needed intervention surface. `induct_skill` stores a reusable function with a description and notes. `retrieve_skill` exposes the stored implementation when the short description is insufficient. Skills accepted after period t become callable from period $t + 1$. This chronological registration preserves the evolving evaluation structure.

The source also gives examples of desirable skill granularity. It encourages compact procedures such as schema-specific loaders and date-to-period converters. Scenario-specific answer dictionaries are discouraged. This contract aligns directly with our targeted temporal-skill design.

4 CONTROLLED SKILL INTERVENTION

The experiment treats a skill as an intervention on the computation available to an agent. The primary question is local: does the procedure associated with a failure improve the same endpoint beyond a matched reusable control?

4.1 FREEZE THE ENDPOINT BEFORE CHANGING THE SKILL

Each endpoint is fixed before arm outcomes are generated. Its benchmark record identifies the requested analytical operation. The same record supplies a deterministic finding verifier. For executable probes, the visible CSV must match its recorded SHA-256. The benchmark reference procedure must also run successfully in an isolated qualification environment. A failed candidate remains in the audit trail and is never replaced after outcomes begin.

4.2 HIDE THE IMPLEMENTATION BEHIND ONE INTERFACE

The executable arms expose the same native tool definition:

```
persistent_skill(): Execute the registered reusable
analysis procedure for the current task when it is
useful.
```

The function name is shared across arms. Its description is shared as well. The JSON argument schema is identical. The model therefore chooses whether to call the tool without seeing which implementation is registered.

When the targeted tool is called, the runner executes the benchmark-recorded procedure on the hash-verified visible data. The original matched control executes neutral schema work through the same tool surface. Its returned text has the same word count as the targeted result. The no-skill arm exposes no additional tool.

The model receives the frozen dialogue through the current user turn. It may answer immediately or issue one native tool call. After the tool returns, the model produces the final answer. A second tool call is a protocol failure. We report tool invocation separately from task success so that an implementation effect cannot be confused with different call rates.

4.3 MEASURE THE DOWNSTREAM ENDPOINT

The final response is scored with the benchmark verifier that existed before our intervention. Keyword endpoints use the frozen benchmark phrases after deterministic normalization. Numerical endpoints require a response value inside the frozen benchmark range. The scorer never receives an arm label.

Let Y_i^T denote success for endpoint i under the targeted implementation. Let Y_i^G denote success under the matched control. The paired effect is

$$\hat{\Delta}_{T-G} = \frac{1}{n} \sum_i (Y_i^T - Y_i^G). \quad (3)$$

The primary test is an exact one-sided paired McNemar/binomial test over discordant endpoints. The preregistered gate requires $\hat{\Delta}_{T-G} \geq 0.20$. It also requires $p \leq 0.05$. Targeted versus no skill is secondary. After execution begins, endpoint membership is fixed. The threshold is fixed as well. Repeated seeds on the same endpoints are not used to alter the primary decision.

4.4 A STRONGER CONTROL FOR PROCEDURE SPECIFICITY

Neutral schema work controls the visible tool interface, yet it leaves a second question: perhaps any substantive analysis would help. We therefore add F6 as a stronger post-hoc diagnostic. F6 reuses the exact F3 targeted initial response for every endpoint. This preserves the model’s original decision about whether to call `persistent_skill`. No new initial model call is made.

For each endpoint, the alternative implementation is another benchmark-recorded analytical procedure from the same TimeSage-MT task. The procedure must execute on the same hash-pinned CSV. Selection is deterministic before F6 continuations: the nearest different procedure in dialogue order is used. Its tool output is matched to the targeted output length. Verifier overlap is recorded and never used for selecting the control.

This diagnostic changes the estimand from *correct procedure versus neutral tool work* to *correct procedure versus another substantive procedure*. Because F6 was designed after observing F3, it has identification value without primary-claim closure authority.

4.5 FROM LOCAL EXECUTABLE VALUE TO PERSISTENT STATE

The TimeSage-MT probes test whether the hidden implementation matters at a frozen endpoint. TimeSage-EV supplies the stronger temporal object. There, a skill must be written before a later release. Its implementation remains fixed when later endpoints arrive. C4 asks whether this state continues to help after the evidence changes. This chronological experiment is distinct from the local implementation swap evaluated here.

Table 2: The six comparisons that determine the paper’s conclusions. “Authority” distinguishes a frozen score-level result from a robustness or directional result.

Comparison	Paired result	Current authority
Persistent targeted text vs. generic text	11/13 vs. 10/13; $p = .500$	no matched signal
Prospective supplied output vs. matched helper	4/20 vs. 5/20; $p = .875$	negative boundary
Executable Qwen vs. neutral matched helper	6/19 vs. 2/19; $p = .109$	directional
First-time STL vs. identical-interface helper	5/7 vs. 0/7; $p = .0313$	deterministic gate pass; semantic robustness open
Non-STL Qwen vs. substantive off-target procedure	5/8 vs. 1/8; $p = .0625$	directional; gate open
Non-STL DeepSeek vs. substantive off-target procedure	5/9 vs. 2/9; $p = .125$	directional; gate open

5 CONTROLLED RESULTS ON TIMESAGE-MT

TimeSage-MT provides public task records with explicit analytical procedures and deterministic downstream checks (Kong et al., 2026). Table 2 shows the comparisons that determine the scientific interpretation. The full execution lineage, including manipulation checks and support-recovery receipts, appears in Appendix B.

5.1 WEAKER SKILL SURFACES DO NOT ISOLATE PROCEDURE VALUE

Persistent text is the weakest intervention. A targeted card describes the benchmark procedure that should be retained after a misleading shortcut. Targeted succeeds on 11/13 endpoints and matched generic text succeeds on 10/13. The paired difference is +7.7 points with $p = 0.50$.

Direct procedure-output injection is stronger because it places analytical results in context. A reachability check confirms that models copy numbers from the supplied output, yet the manipulation endpoint is not independent of that output. We therefore freeze 20 new tasks and score the benchmark’s downstream finding. Targeted output succeeds on 4/20 tasks while the matched helper succeeds on 5/20. The paired difference is −5.0 points with $p = 0.875$. Making the correct procedure result available in context does not establish matched downstream repair under this realization.

5.2 EXECUTABLE STATE PRODUCES THE FIRST MATCHED GATE PASS

The executable intervention hides the arm behind one tool surface. The target and matched arms expose the same function name. Their descriptions and argument schemas are identical. The model chooses whether to call the tool before it sees the hidden implementation result.

In the initial Qwen experiment, 19 endpoints pass pre-outcome execution qualification. Targeted succeeds on 6/19 endpoints and a content-light matched executable helper succeeds on 2/19. The paired effect is +21.1 points. Five discordant endpoints favor targeted and one favors the helper, giving exact one-sided $p = 0.109375$. The preregistered +20-point magnitude criterion is met, while the full matched gate stays open. DeepSeek-v4-Pro preserves the positive direction on the same endpoints at 5/19 versus 3/19.

A separate candidate population was frozen before any later support recovery. Seven genuinely first-time recovered endpoints form an independent tranche after previously evaluated tasks are removed. Targeted succeeds on 5/7 endpoints while matched generic succeeds on 0/7. All five discordant pairs favor targeted, giving +71.4 points with exact one-sided $p = 0.03125$. This passes the frozen deterministic matched gate. Every endpoint in this tranche is STL decomposition, so the formal score-level support is procedure-family specific.

We audit all 21 answers from these seven endpoints with arm labels hidden. One of the two predeclared semantic reviewers returns a complete judgment set; the other is nonvoting after a function-argument parse failure and is not replaced. The valid reviewer judges 6/7 targeted answers and 2/7 generic answers fully correct. Five pairs favor targeted and one favors generic, giving +57.1 points with exact

$p = 0.109375$. The semantic audit therefore does not reproduce the deterministic matched gate. It leaves the frozen score unchanged while making measurement robustness an explicit evidence debt.

The original frozen population can also be summarized after later support completion using exactly one first valid observation per executable endpoint. Twenty endpoints are available. Targeted succeeds on 10/20 and matched generic on 2/20, giving a descriptive +40.0-point effect with $p = 0.00390625$. Because part of the support completion occurs after partial outcomes, this combined statistic is treated as robustness evidence and does not create a second closure test.

5.3 A SUBSTANTIVE OFF-TARGET CONTROL TESTS BROADER PROCEDURE FAMILIES

The initial executable control performs real tool invocation while carrying little analytical content. We therefore build a new strong-control population from TimeSage-MT L2. Its 60 task IDs are frozen as a source population before any new model outcome. Fifty-eight task JSONs are available under the bounded support snapshot; two remain support debt. Measurement qualification selects the earliest non-basic target procedure whose benchmark reference output is finite and consistent with its deterministic verifier. The task must also contain an earlier substantive benchmark procedure that does not satisfy the target verifier. Twelve tasks meet the metadata rule. Nine pass execution qualification. One remains support hold and two fail reference replay or execution. No replacement task is added.

The nine qualified endpoints span five non-STL target families: time-series forest regression, residual diagnostics, linear interpolation, model comparison and ROCKET classification. The targeted tool executes the target benchmark procedure. The strong control executes the selected task-local off-target procedure. Both outputs are truncated to the same pre-frozen word count. The visible tool definition is identical across the two executable arms.

On Qwen2.5-32B, one off-target arm attempts a forbidden second tool call. The corresponding endpoint is invalid under the frozen protocol and is not retried. Eight paired endpoints remain. Targeted succeeds on 5/8 and the substantive off-target control on 1/8. The paired effect is +50.0 points. Four discordant pairs favor targeted and none favor the control, giving exact one-sided $p = 0.0625$. This reaches the magnitude criterion and misses the inferential threshold by one exact step.

We repeat the exact nine-endpoint contract with DeepSeek-v4-Pro. All 27 arm records are valid. Targeted succeeds on 5/9 endpoints and the substantive off-target control succeeds on 2/9. The paired effect is +33.3 points with $p = 0.125$. No-skill succeeds on 0/9, giving a secondary targeted-no-skill $p = 0.03125$. Qwen and DeepSeek are adjudicated independently; their p -values are never pooled.

These strong-control experiments broaden the evidence beyond STL decomposition. The targeted-minus-off-target direction is positive on both model families and across five non-STL target procedures. Neither independent broad-family matched gate closes. We therefore retain broad C3 as active. The STL tranche remains the only frozen deterministic gate pass, and F11 shows that its semantic measurement robustness is not established.

6 RELATED WORK

Evolving time-series agents. TimeSage-EV evaluates repeated analysis under changing release periods and strict evidence cutoffs (Yao et al., 2026). Its TimeSage-1.0 harness adds reusable skills that become available to later periods. TimeSage-MT provides multi-turn tasks with explicit analytical skill requirements and verifiable task records (Kong et al., 2026). We use TimeSage-EV to define the longitudinal scientific object and TimeSage-MT to isolate intervention surfaces under matched controls.

Experience-derived agent memory. Agent Workflow Memory induces reusable procedures from prior trajectories (Wang et al., 2024). ReasoningBank stores reasoning memories derived from experience (Ouyang et al., 2026). MUSE-Autoskill studies skill creation together with later management and evaluation (Lin et al., 2026). These systems establish persistent procedural state as a useful design pattern. Our question concerns causal credit: which part of a stored skill changes later behavior after generic assistance is matched?

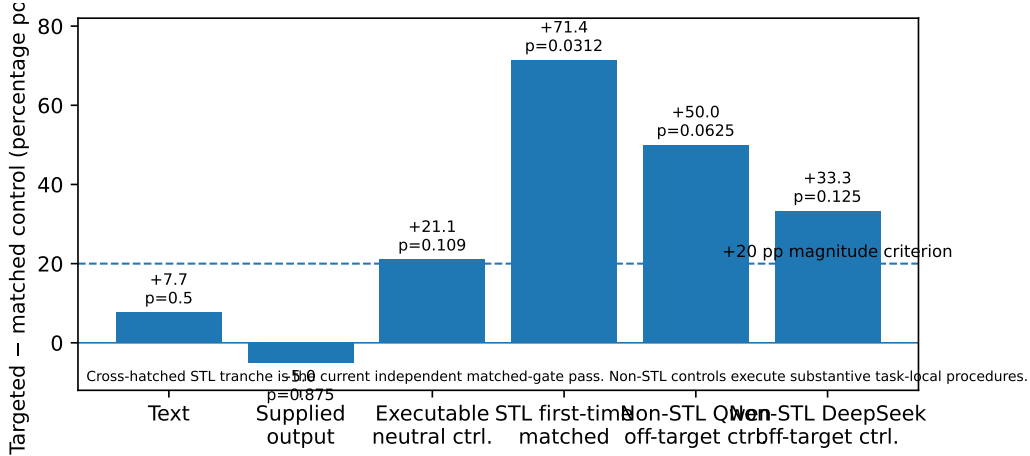


Figure 3: Matched targeted-minus-control effects for the intervention surfaces that determine the paper’s conclusions. The first STL tranche passes its frozen matched gate. The non-STL strong-control experiments are directionally positive on Qwen and DeepSeek while their exact matched gates remain open.

Validated skill evolution. SkillCommit emphasizes behaviorally validated scope expansion before a skill is committed to broader reuse (He & Yang, 2026). Work on dynamic skill libraries similarly highlights the need to evaluate skills after creation (Li, 2026). Our protocol adds a matched implementation swap. The visible tool interface stays fixed while the hidden executable procedure changes. This design separates interface prompting from implementation value.

Skill libraries as intervention surfaces. A modular skill library exposes a causal boundary that ordinary conversation memory does not. The same model can receive the same dialogue while one registered procedure changes. F3 operationalizes this boundary with a native tool loop. The TimeSage chronological contract then extends the same idea across release periods, where a skill written earlier becomes state for a later endpoint.

7 LIMITATIONS

The strongest unresolved support gap is the TimeSage-EV evaluated snapshot. The source paper declares a public release, yet the May 2026 period-sequential artifacts are absent from our auditable frozen support set. This prevents the chronological replay in which a skill is written before a later release and remains available as persistent state. Repository access failures have no scientific authority against C4.

The executable interventions supply benchmark-recorded reference procedures. Skill induction is outside the intervention. The experiments estimate downstream value after a candidate procedure is registered. They do not establish that an agent can discover the same procedure from its own earlier failure.

The first formal matched-gate pass is concentrated in STL decomposition. F8 contains seven first-time STL endpoints and gives 5/7 targeted successes versus 0/7 generic successes. A deduplicated first-observation completion of the pre-outcome frozen population gives 10/20 versus 2/20 with $p = 0.00390625$. Part of that support completion occurs after partial outcomes, so the combined statistic is robustness evidence and does not create a second formal closure test. One original candidate remains support hold. Three recovered candidates fail reference execution or replay. No failed candidate is replaced.

The new L2 strong-control population broadens the procedure families and strengthens the comparator. Nine endpoints pass execution qualification across five non-STL target procedures. On Qwen, one endpoint becomes invalid because the off-target arm issues a second tool call forbidden by the frozen protocol. The remaining targeted-off-target comparison is 5/8 versus 1/8 with $p = 0.0625$. DeepSeek

gives 5/9 versus 2/9 with $p = 0.125$. The point-estimate direction is positive in both model families, while neither broad-family primary gate closes. Broad C3 therefore remains active.

The strong off-target controls are substantive procedures from the same benchmark task, yet they are still chosen from benchmark-recorded analyses. They provide a stronger alternative than neutral schema work without covering every plausible wrong procedure an evolving agent could store.

The benchmark verifier is deterministic but coarse. Keyword checks can miss a valid paraphrase. Numerical-range checks can accept an isolated in-range value without validating the full explanation. An earlier semantic audit produced zero voting judgments because both provider calls were incomplete. We later audit all 21 F8 answers with arm labels hidden. One predeclared reviewer returns valid judgments and the second is nonvoting after a parse failure; no replacement reviewer is added after labels are observed. The valid reviewer gives 6/7 targeted and 2/7 generic answers strict semantic credit, yielding $p = 0.109375$ for the matched comparison. The frozen deterministic F8 score remains unchanged, while semantic measurement robustness is not established.

Two model families are evaluated for the main executable patterns. This remains a narrow basis for a claim about agent models in general. The non-STL strong-control replication reduces single-model dependence, while more model families would still strengthen external validity.

We stop same-substrate experiment expansion after a pre-outcome source audit. The unused L2 source contains zero measurement-valid fresh STL confirmation tasks after excluding the F9 candidate pool. A complete 60-task L1 snapshot yields only 12 fresh task-level strong-control candidates across stationarity, seasonality, autocorrelation and cross-correlation procedures. This is below the predeclared minimum of 16 fresh pairs required to authorize another experiment. No F12 model outcome is generated, and the minimum is not relaxed after observing the source count.

C4 requires release-to-release persistence. None of the TimeSage-MT probes contains a skill-write event followed by later-period reuse. The longitudinal claim remains experiment debt until the TimeSage-EV evaluated artifacts, or an equivalent first-party period-sequential substrate, become auditable.

8 CONCLUSION

A reusable agent skill is not one experimental object. Persistent text changes what the model reads. Supplied procedure output changes what information is placed in context. Executable state changes what computation becomes available after the model chooses to call a tool.

Our controlled experiments separate these surfaces. The weaker text and output interventions do not show matched downstream gains. Executable state produces the strongest matched effects. An independent first-time STL tranche passes the frozen deterministic targeted-generic gate at 5/7 versus 0/7 with exact $p = 0.03125$. A blinded semantic diagnostic does not reproduce that inferential gate with its only valid reviewer, so measurement robustness remains open. A 20-endpoint first-observation completion supplies additional score-level robustness without changing the tranche-level authority.

A new L2 population tests five non-STL target procedure families against substantive task-local off-target procedures. Qwen gives 5/8 targeted successes versus 1/8 off-target success. DeepSeek gives 5/9 versus 2/9. Both matched directions are positive, while the exact primary gates remain open. The current evidence therefore shows a frozen deterministic executable effect within STL decomposition. Semantic robustness remains unresolved. The stronger causal control provides broader directional evidence.

The resulting evaluation standard is simple: hold the visible callable interface fixed and test the hidden implementation against a substantive matched alternative. A longitudinal self-evolution claim additionally needs the state transition that writes the procedure before later evidence arrives.

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A TIMESAGE-EV SKILL CONTRACTS

Temporal cutoff. The skill receives a target period and candidate evidence timestamps. It returns the evidence that is admissible under the declared cutoff. The procedure contains no scenario answer.

Release alignment. The skill receives source-release metadata and returns the analytical period represented by that release. It also checks the selected time-series row against the normalized period key.

Exogenous grounding. The skill receives a target period and retrieves cutoff-valid documentary evidence for an attribution. It returns evidence identifiers for later reasoning.

Matched callable control. A faithful TimeSage-EV control uses the same function surface as the targeted skill. The execution budget is fixed before outcomes. The control cannot inspect the target failure label or a scenario-specific answer.

B INDEPENDENT PROBE PROTOCOLS

B.1 F0 PERSISTENT-TEXT PROBE

F0 contains 13 TimeSage-MT correction endpoints and 39 deterministic model calls. Endpoint identities are frozen before arm execution. Targeted succeeds on 11/13 endpoints. Generic succeeds on 10/13. No skill succeeds on 8/13. The targeted-generic paired difference is 1/13 with one-sided exact $p = 0.50$.

B.2 F1 REACHABILITY MANIPULATION CHECK

F1 contains 24 endpoints that are disjoint from F0. The targeted arm receives benchmark-recorded procedure output. The generic arm receives a length-matched metadata output. Three numeric anchors are frozen from the target output before arm execution. Reproducing two anchors counts as manipulation success. Targeted succeeds on 14/24 endpoints while generic succeeds on 1/24. This check verifies that procedure output reaches the response.

A deterministic downstream diagnostic uses the benchmark finding metadata on the unchanged F1 responses. Thirteen endpoints have compatible keyword or numerical-range verifiers. Targeted succeeds on 6/13 and generic on 5/13, giving a +7.7-point difference with $p = 0.50$.

B.3 F2 PROSPECTIVE OUTPUT-INJECTION PROBE

F2 freezes 20 new task-disjoint endpoints before model calls. The primary endpoint is the benchmark downstream finding. Targeted succeeds on 4/20. Generic succeeds on 5/20. No skill succeeds on 6/20. The targeted–generic difference is -0.05 with $p = 0.875$. The frozen gate fails.

C F3 EXECUTABLE SKILL-STATE PROTOCOL

F3 begins from 25 unused TimeSage-MT tasks. Each candidate has one deterministic benchmark verifier and one recorded reference procedure from the allowed temporal-skill set. The earliest eligible turn is selected before any F3 model output.

Execution qualification is performed before model evaluation. The public visible CSV must match the task’s recorded SHA-256. The benchmark reference code must terminate successfully and emit a finite result. Nineteen tasks pass. Two tasks fail execution qualification. Four tasks have support failures. No failed task is replaced.

The qualified set contains seven `stl_decompose` endpoints. Four use `cross_correlation`. Four use `rolling_stat`. Four use `seasonality_detect`. Eleven tasks are L3 and eight are L4.

C.1 MODEL-BLIND IMPLEMENTATION SWAP

Targeted and generic arms expose this same native tool definition:

```
persistent_skill(): Execute the registered reusable
analysis procedure for the current task when it is
useful.
```

The function name is identical. The description is identical. The argument schema is identical. The model receives no arm label.

The targeted implementation executes the frozen benchmark reference code on the hash-verified CSV at call time. Its returned result is capped at 80 words. The generic implementation reads the same CSV and performs neutral schema validation. It then returns neutral text with exactly the same word count as the targeted result. Frozen benchmark verifier keywords are absent from the generic output.

The model first sees the frozen dialogue and the registered tool. It may answer directly or issue a native tool call. After one tool result, the model produces the final response. A second tool call is a protocol failure. Formal execution completes all 57 arm records with zero support failures and zero protocol failures.

Targeted calls the tool on 17/19 endpoints. Generic also calls it on 17/19. Targeted succeeds on 6/19. Generic succeeds on 2/19. No skill succeeds on 0/19. The primary comparison has five targeted-only successes and one generic-only success. This gives $\hat{\Delta}_{T-G} = 4/19 = 0.2105$ and one-sided exact $p = 0.109375$. The secondary no-skill comparison has six targeted-only successes and gives $p = 0.015625$.

The predeclared skill rows are descriptive. STL yields 5/7 targeted successes and 1/7 generic success. Rolling statistics yields 1/4 in both executable arms. The cross-correlation row has zero successes in all arms. The seasonality row also has zero successes in all arms.

D F4 CROSS-MODEL EXECUTABLE REPLICATION

F4 reuses the exact 19 F3 endpoints. The visible CSV hashes and reference-code outputs are requalified before any F4 model outcome; all 19 pass. The model changes to DeepSeek-v4-Pro. Temperature is fixed at zero. The tool definition and downstream verifier are unchanged from F3.

Targeted succeeds on 5/19 endpoints. Generic succeeds on 3/19. No skill succeeds on 0/19. The targeted–generic comparison has three targeted-only successes and one generic-only success. Its paired difference is $2/19 = 0.1053$ with one-sided exact $p = 0.3125$. The targeted–no-skill

comparison has five targeted-only successes and gives $p = 0.03125$. F4 is adjudicated independently from F3, and their p -values are never pooled.

The targeted tool is called on 17/19 endpoints and the generic tool on 18/19. Three endpoints differ in invocation choice. The generic arm therefore receives at least as many observed tool invocations in aggregate. F4 preserves the positive targeted–generic direction without closing the primary matched gate.

E F5 ALL-REMAINING PUBLIC-SUPPORT REPLICATION

F5 excludes every F0–F2 task and every task that appeared in the F3 pre-outcome candidate manifest. The remaining metadata scan freezes 24 candidate tasks before any F5 model output. Qualification admits every task whose public visible CSV can be hash-validated and whose benchmark reference code reproduces the recorded output. Seven tasks pass. Seventeen are recorded as support failures and no replacement task is added.

The seven endpoints contain five STL-decomposition tasks and two stationarity-test tasks. All are L3 tasks. F5 uses the same Qwen2.5-32B tool surface and downstream verifier as F3. Targeted succeeds on 3/7 endpoints. Generic succeeds on 2/7. No skill succeeds on 0/7. The primary difference is $1/7 = 0.1429$ with one targeted-only discordance and exact $p = 0.50$. The secondary targeted–no-skill difference is $3/7 = 0.4286$ with $p = 0.125$.

The first formal execution contains a local path error: both executable arms make the frozen initial tool-call decision, then fail before local tool execution because the mechanical F3 runner points at the wrong cache directory. It yields zero valid paired endpoints. Runtime repair R1 keeps the F5 contract unchanged. It reuses each original targeted or generic initial model response. The hash-bound tool then executes from the cache used during qualification. The repair sends only the required final continuation. Original no-skill responses are reused without a second model sample. The repair makes zero new initial model calls. All 21 repaired arm records complete with zero support failures and zero protocol failures.

F5 is adjudicated independently. Its exact test is never pooled with F3 or F4. The 17 qualification support failures remain visible as support debt and have no scientific authority.

E.1 SUPPORT-RECOVERED F5-R2 EXTENSION

A later support-only recovery revisits the same 17 frozen F5 support failures. Candidate membership is unchanged. Five exact visible CSVs are recovered from existing local cache by SHA-256; twelve candidates remain unavailable. The five recovered candidates are requalified with the original reference-code rule and all five pass. No new candidate is introduced.

F5-R2 uses the unchanged Qwen tool protocol and is adjudicated separately. Targeted succeeds on 1/5 endpoints. Generic succeeds on 0/5. No skill succeeds on 1/5. The targeted–generic difference is $1/5 = 0.20$ with one discordant pair and exact $p = 0.50$. Targeted versus no skill has zero paired difference. This extension is never pooled with F5 or the earlier executable probes.

F F8 DEDUPLICATED FROZEN-POPULATION SUPPORT RECOVERY

The original F5 candidate manifest contains 24 tasks and was frozen before any F5 model outcome. After F5-R2, twelve additional support-debt candidates become executable through exact-SHA recovery. The recovery adds no candidate and does not use model outcomes to determine membership. Reference code is replayed on each recovered CSV before arm execution. All twelve recovered candidates pass the original execution rule.

Five of those twelve tasks had already been evaluated in F5-R2. They are L3_energy_045, L3_retail_022, L4_climate_048, L4_telecom_036 and L4_telecom_055. Their repeated calls receive zero new scientific authority. F8 is defined on seven first-time tasks only. They are L3_climate_018, L3_finance_052, L3_healthcare_049, L4_climate_020, L4_climate_053, L4_finance_002 and L4_healthcare_044.

The seven first-time endpoints all use `stl_decompose`. Targeted succeeds on 5/7. Matched generic succeeds on 0/7. No skill succeeds on 1/7. The primary targeted–generic comparison has five targeted-only successes and no generic-only success. This gives $\hat{\Delta}_{T-G} = 5/7 = 0.7143$ with one-sided exact $p = 0.03125$. The frozen +0.20 and $p \leq 0.05$ gate passes. The secondary targeted–no-skill comparison has four targeted-only successes and gives $p = 0.0625$.

The full twelve-endpoint rerun gives 6/12 targeted successes and 0/12 generic successes, yet five rows repeat F5-R2. That twelve-row statistic is retained only as a deterministic rerun diagnostic. A separate deduplication adjudication records the seven rows with first-time scientific authority.

F.1 FIRST-OBSERVATION COMPLETION OF THE FROZEN F5 POPULATION

A later support-only pass resolves additional exact-SHA inputs without adding candidates. One new endpoint, `L3_healthcare_010`, receives its first model evaluation. Twelve other newly called endpoints had already been evaluated in earlier F5-R2 or F8 tranches; those repeated calls carry zero new scientific authority.

Using the first valid observation for each execution-qualified member of the original 24-task F5 manifest gives 20 endpoints. Targeted succeeds on 10/20. Generic succeeds on 2/20. No skill succeeds on 2/20. The targeted–generic comparison has eight targeted-only successes and no generic-only successes. This gives $\hat{\Delta}_{T-G} = 8/20 = 0.40$ with one-sided exact $p = 0.00390625$. The targeted–no-skill comparison has the same paired difference and exact p -value.

This 20-endpoint statistic is recorded as descriptive robustness evidence. It does not receive new formal closure authority because the support-completion operation occurs after partial F5 outcomes. One candidate remains support hold. Three recovered candidates fail reference execution or replay. None is replaced.

G F6 STRONGER SUBSTANTIVE-CONTROL DIAGNOSTIC

F6 addresses the possibility that neutral schema validation is too weak a matched control. It reuses the exact 19 F3 endpoints. For each task, qualification searches only benchmark-recorded agent procedures from that same task. The nearest different procedure in dialogue order is selected deterministically. The alternative must execute on the same hash-pinned visible CSV. All 19 endpoints obtain a valid alternative procedure.

F6 makes zero new initial model calls. It reuses the original F3 targeted initial response for each endpoint, including the original decision about whether to call `persistent_skill`. Seventeen endpoints had called the tool and therefore receive one new final continuation after the alternative tool result. The two no-tool endpoints reuse their original final response. Tool-output word count is matched to the F3 targeted output. Verifier overlap is recorded and is never used to choose the alternative.

The original correct F3 procedure succeeds on 6/19 endpoints. The substantive off-target procedure succeeds on 1/19. There are six targeted-only successes and one off-target-only success. The paired difference is $5/19 = 0.2632$ with one-sided exact $p = 0.0625$. Because F6 was designed after observing F3, its result is a diagnostic of control strength and is excluded from primary C3 closure.

H F9–F10 NON-STL STRONG-CONTROL REPLICATION

F9 draws from a new TimeSage-MT L2 source population. All 60 L2 task IDs are frozen before any F9 model output. Fifty-eight task JSONs are retrieved and content-addressed under the bounded snapshot; two remain support hold. Candidate construction excludes basic profiling procedures as targets. A target must have benchmark reference code, finite reference output and a deterministic verifier that is satisfied by that reference output. Each task must also contain an earlier substantive benchmark procedure whose output does not satisfy the target verifier.

Twelve tasks meet the metadata rule. Execution qualification requires both the target and off-target reference procedures to run on the same hash-verified visible CSV and reproduce their benchmark outputs. Nine tasks pass. One visible-data item remains support

hold. Two tasks fail reference execution or replay. No replacement task is added. The qualified target procedures cover `ts_forest_regressor_tsr`, `residual_diagnostics`, `linear_interpolation`, `model_comparison` and `rocket_classify`.

Target and off-target arms expose the same `persistent_skill` definition. The target implementation executes the qualified target procedure. The control executes the qualified prior procedure from the same task. Both tool outputs are truncated to the same pre-frozen word count. The no-skill arm has no registered tool. The target endpoint keeps its original benchmark verifier.

F9 uses Qwen2.5-32B. One off-target response issues a forbidden second tool call after the first tool result, so that endpoint is invalid under the frozen protocol and is never retried. Eight paired endpoints remain. Targeted succeeds on 5/8. Off-target succeeds on 1/8. The primary comparison has four targeted-only successes and no off-target-only success, giving $\hat{\Delta} = 4/8 = 0.50$ and exact one-sided $p = 0.0625$. No skill succeeds on 2/8.

F10 repeats the exact nine-endpoint F9 contract with DeepSeek-v4-Pro. All 27 arm records are valid. Targeted succeeds on 5/9, off-target on 2/9 and no skill on 0/9. The targeted-off-target comparison has three targeted-only successes and no off-target-only success, giving $\hat{\Delta} = 3/9 = 0.3333$ with $p = 0.125$. Targeted versus no skill gives $p = 0.03125$. F9 and F10 are adjudicated separately; their p -values are never pooled.

F9 candidate-manifest SHA-256 begins `0c76d3a010ac`. Its qualification begins `ecd8a136b69b`. The F9 contract begins `f813ba55292c`. Its runner begins `91578fa7b005`. Its result begins `6a84f84097e3`. The F10 contract begins `7bf59505d197`. Its runner begins `07fd1b1bc7eb`. Its result begins `979515627a1c`. Independent recomputation verifies all 27 raw files for each model and reports zero integrity issues.

I F11 BLINDED SEMANTIC MEASUREMENT AUDIT

F11 audits every answer from the seven first-time F8 endpoints. All three arms are included, giving 21 answers. Arm labels are removed before review and answer order is deterministically shuffled. The reviewer sees the benchmark dialogue, key finding, deterministic verifier metadata, reference stdout and one anonymous answer at a time. The diagnostic success rule is strict semantic CORRECT; PARTIAL does not count as success. The frozen F8 deterministic scores and gate are never changed by this audit.

Two semantic reviewers are predeclared. DeepSeek-v4-Pro returns a complete 21-answer judgment set. It assigns strict correctness to 6/7 targeted answers, 2/7 generic answers and 0/7 no-skill answers. Five matched pairs favor targeted and one favors generic, so the targeted-generic semantic difference is $4/7 = 0.5714$ with one-sided exact $p = 0.109375$. Counting PARTIAL as correct in a sensitivity calculation gives 6/7 targeted and 3/7 generic, with $p = 0.1875$.

The second reviewer, Kimi-K3, returns malformed function arguments that cannot be parsed into a complete auditable judgment set. It is therefore nonvoting. No replacement reviewer is added after the DeepSeek labels are observed. F11 consequently cannot establish semantic robustness. It records that the frozen deterministic F8 gate pass is measurement-sensitive under the available blinded audit.

The deterministic and semantic scorers disagree on several answers. DeepSeek semantically accepts two targeted answers that the deterministic verifier misses, rejects one targeted deterministic success, accepts two generic answers that the deterministic verifier misses and rejects the only deterministic no-skill success. These discrepancies motivate keeping the semantic measurement question open without changing the frozen benchmark score.

J REPRODUCIBILITY AND ARTIFACT BINDING

The claim ledger binds the public probe-summary artifact by SHA-256. The numeric reproduction entrypoint is `reproduce_d2_temporal_skill_paper.py`. Passing `--figure` regenerates Figure 3.

F1-v2 has contract SHA-256 1c84fd55226b...3e2d6. Its result SHA-256 is ef37e518a2fc...febb1. The final F2 contract SHA-256 is 6fc6e4d7b3a6...a59c6. The F2 result SHA-256 is 4ab9ad27639f...dfb3.

F3 candidate manifest SHA-256 is 8619d4a099e6...b155d. The qualification artifact SHA-256 is 37f8006794be...b9599. The frozen F3 contract SHA-256 is 494636b47b48...74d7. The runner SHA-256 is ffd880180621...9444f. The result SHA-256 is ace9d116ce29...9fdb. An independent recomputation reads all 57 F3 response files and verifies 57 unique raw hashes. It reproduces the arm counts and exact paired statistics.

F4 requalification SHA-256 is 1c29758d4db4...878c. The frozen F4 contract SHA-256 is 5cfeed24dea1...a873. The runner SHA-256 is 3f010c126ad8...6668. The result SHA-256 is 7419c59818b2...8851. Its 57 response files have unique hashes, and independent recomputation reproduces the reported DeepSeek arm counts and exact tests.

F5 candidate-manifest SHA-256 is b49cd462bda8...c03e. Qualification SHA-256 is eb104c294d2f...163a1. The frozen F5 contract SHA-256 is c8ae6e0de7ae...6399. The initial failed result SHA-256 is 0e0e5e49fb12...f02a. Runtime-repair receipt SHA-256 is 5f1b0f64de03...c6bb. The repaired result SHA-256 is 4a3456e952f2...6f9b. Independent recomputation reads 21 repaired response records and reproduces the arm counts.

F5 support-recovery SHA-256 is a09e73b65a3c...7dee6. F5-R2 requalification SHA-256 is 2a2e57d68fe1...9dc1. Its frozen contract SHA-256 is bcf53dc65276...ff75; result SHA-256 is afb8893d0507...3c10.

F8 uses the later exact-SHA recovery snapshot c4ded18a4e7b...99bc8. Its qualification SHA-256 begins 153ce397ef23. The contract begins 5b7fdabfad11. The runner begins 49614d390d58. The twelve-row result begins 202eb90926. Deduplication adjudication is published at generated/d2-temporal-skill-f8-dedup-adjudication-20260822.json. It assigns independent authority to seven first-time endpoints and zero new authority to the five repeated F5-R2 tasks.

The later first-observation completion is published in the artifact bundle with SHA-256 prefix 953cd2b50ba6. It adds one first-time endpoint and reports the 20-endpoint deduplicated robustness summary while preserving zero new authority for repeated R5 calls.

F6 qualification SHA-256 is f0c7cca0286c...8181. The frozen diagnostic contract SHA-256 is 114b4d9a76d8...5e4a. Runner SHA-256 is e71504f5af5a...2d5. Result SHA-256 is aadd01dc7d0a...7db. Independent recomputation reads all 19 F6 response files and reproduces the paired statistic.

The public F11 artifact has SHA-256 prefix e1c02b624344. The frozen semantic-audit contract begins 2ba57d779f38. The valid DeepSeek judgment receipt begins 95f84d7c4dc9. The nonvoting Kimi parse-failure receipt begins 2751b39bb3fa. No provider response identifier is published.

F0–F3 use Qwen2.5-32B. F5, F6, F8 and F9 also use Qwen2.5-32B. F4 and F10 use DeepSeek-v4-Pro. F11 uses blinded semantic review and does not alter any model-arm output. The executable probes use temperature zero through a registered tool surface whose visible definition is matched within each experiment. The public artifact omits server paths and provider-local runtime details.

K FRESH-SOURCE SATURATION AND FROZEN TIMESAGE-EV SUPPORT STATE

A final pre-outcome source audit asks whether another same-substrate confirmation experiment is justified. The GO rule requires at least 16 fresh measurement-valid task-level pairs before any new model outcome. In the unused L2 source, no fresh task supports an STL target with a self-consistent deterministic verifier after excluding the F9 candidate pool. In L1, all 60 task JSONs are content-addressed, yet only 12 task-level target/control pairs satisfy the metadata rule. They cover stationarity testing, seasonality detection, ACF/PACF and cross-correlation. The source count is below the frozen

GO threshold, so no F12 model experiment is launched. The threshold is not relaxed after observing this count.

The public source-saturation artifact has SHA-256 prefix 194a2e5a0621. Its private adjudication begins f02f84ef3221. The L1 source snapshot begins 0344a59200e1; its candidate manifest begins adfba197008c. The artifact forbids same-endpoint seed extension, endpoint reuse as independent confirmation and multiple correlated turns from one task used to inflate effective sample size.

The locally audited TimeSage-EV first-party release does not contain the evaluated May 2026 period-sequential snapshot required for the faithful replay. As of August 23, 2026, the public TimeSage-Series/TimeSage-EV repository exposes a README marked “Upcoming” without the evaluated snapshot. This support state prevents the exact C4 test and carries no negative scientific authority. The TimeSage-MT probes remain separately identified from the TimeSage-EV longitudinal claim in the manuscript and claim ledger.